

UNIT 1

- ASP.NET – Introduction, overview of ASP.NET framework
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Introduction

- ASP.NET is a web development platform, which provides a programming model, a comprehensive software infrastructure and various services required to build up robust web applications for PC, as well as mobile devices.
- ASP.NET works on top of the HTTP protocol, and uses the HTTP commands and policies to set a browser-to-server bilateral communication and cooperation.
- ASP.NET is a part of Microsoft .Net platform. ASP.NET applications are compiled codes, written using the extensible and reusable components or objects present in .Net framework. These codes can use the entire hierarchy of classes in .Net framework.
- The ASP.NET application codes can be written in any of the following languages:
 - C#
 - Visual Basic. Net
- - Jscript
 - J#
- ASP.NET is used to produce interactive, data-driven web applications over the internet. It consists of a large number of controls such as text boxes, buttons, and labels for assembling, configuring, and manipulating code to create HTML pages.

Overview of ASP.NET framework

- The .NET Framework is a technology that supports building and running the next generation of apps and XML Web services. The .NET Framework is designed to fulfill the following objectives:
- To provide a consistent object-oriented programming environment whether object code is stored and executed locally, executed locally but Internet-distributed, or executed remotely.
- To provide a code-execution environment that minimizes software deployment and versioning conflicts.
- To provide a code-execution environment that promotes safe execution of code, including code created by an unknown or semi- trusted third party.
- To provide a code-execution environment that eliminates the performance problems of scripted or interpreted environments.

- To make the developer experience consistent across widely varying types of apps, such as Windows-based apps and Web-based apps.
- To build all communication on industry standards to ensure that code based on the .NET Framework integrates with any other code.
- Active Server Pages (ASP) is a server-side scripting technology.
- ASP is a technology that Microsoft created to ease the development of interactive Web applications. With ASP you can use client-side scripts as well as server-side scripts. Maybe you want to validate user input or access a database. ASP provides solutions for transaction processing and managing session state.
- Asp is one of the most successful language used in web development.

Understanding ASP.NET Controls

- The ASP.NET Framework contains more than 90 controls.
- These controls can be divided into seven groups:

Standard Controls

- Enable you to render standard form elements such as buttons, input fields, and labels.

Validation Controls

- Enable you to validate form data before you submit the data to the server. For example, you can use a RequiredFieldValidator control to check whether a user entered a value for a required input field.

Rich Controls

- Enable you to render things such as calendars, file upload buttons, rotating banner advertisements, and multistep wizards.

Data Controls

- Enable you to work with data such as database data. For example, you can use these controls to submit new records to a database table or display a list of database records.

Navigation Controls

- Enable you to display standard navigation elements such as menus, tree views, and bread crumb trails.

Login Controls

- Enables you to display login, change password, and registration forms.

HTML Controls

- Enable you to convert any HTML tag into a server-side control.

Understanding ASP.NET pages

When adding a new ASP.NET page to your website, you can instruct Visual Studio to put the source code portion in one of two locations:

. In a separate file

- The ASP.NET page is composed of two separate files,

PageName.aspx and PageName.aspx.vb. The PageName.aspx file contains The HTML portion whereas the PageName.aspx.vb file contains the source code.

This is how our current sample web page is structured, with Default.aspx and Default.aspx.vb.

. In a <script> block in the same file

- The HTML and source code portions can reside in the same file. In this scenario, the source code portion is placed within a <script runat =”server”> element.

Advantages of ASP.NET

1. Language Independence:

ASP.NET supports multiple programming languages like C#, VB.NET, JScript.NET, and others. This allows developers to choose the language they are most comfortable with or that best suits the project requirements.

2. Server-side Technology:

ASP.NET is a server-side technology, which means that the code is executed on the server before it is sent to the client's browser. This can lead to better performance and improved security.

3. Rapid Development:

ASP.NET includes a variety of tools and libraries that facilitate rapid application development. Features like drag-and-drop server controls, automatic deployment, and integrated development environment (IDE) support in Visual Studio contribute to faster development cycles.

4. Scalability:

ASP.NET applications are highly scalable. They can handle a large number of users and can be easily scaled up by adding more servers to the server farm. This makes ASP.NET suitable for both small websites and large enterprise-level applications.

5. Security Features:

ASP.NET incorporates a number of security features, including code access security, encryption, and authentication options. It helps developers build secure applications and protects against common web vulnerabilities.

6. Rich Class Library:

ASP.NET provides a rich class library that includes a wide range of pre-built functions and libraries. This reduces the amount of code developers need to write for common tasks and promotes code reusability.

7. ASP.NET MVC Framework:

The ASP.NET MVC (Model-View-Controller) framework allows developers to build modular and maintainable web applications by separating concerns. It promotes a clean and organized code structure, making it easier to manage and extend applications.

8. State Management:

ASP.NET provides various options for state management, including server-side and client-side options. This flexibility allows developers to choose the best approach based on the specific requirements of their application.

9. Integration with Other Microsoft Tools:

ASP.NET integrates well with other Microsoft technologies and tools, such as SQL Server, Azure, and Visual Studio. This seamless integration simplifies the development process and enhances productivity.

10. Community Support:

ASP.NET has a large and active community of developers. This community support provides access to a wealth of resources, tutorials, and third-party libraries, making it easier for developers to find solutions to common challenges.

Web Server

Web server is a computer where the web content is stored. Basically web server is used to host the web sites but there exists other web servers also such as gaming, storage, FTP, email etc.

Web Server Working

Web server respond to the client request in either of the following two ways:

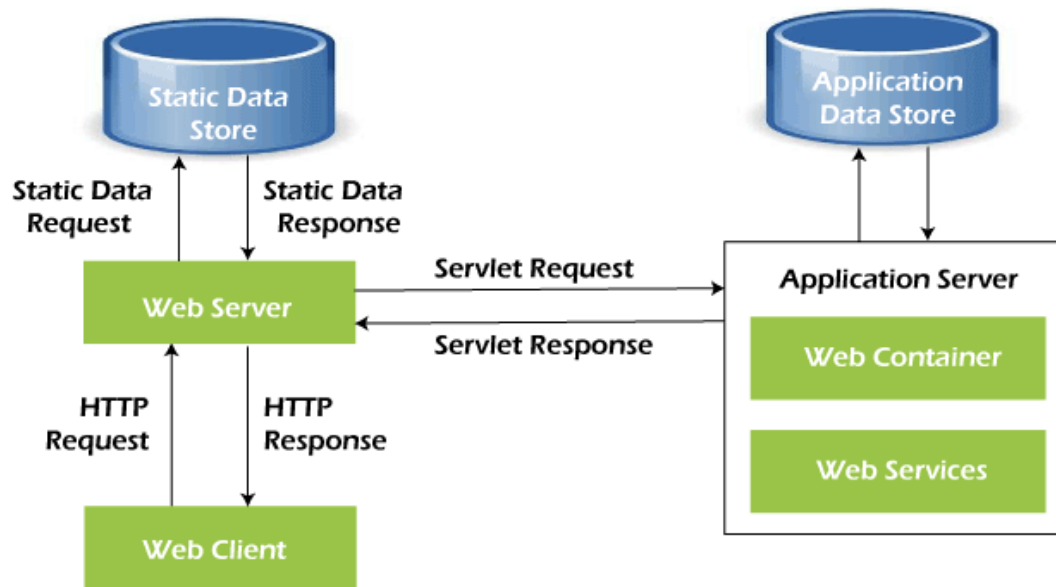
Sending the file to the client associated with the requested URL. Generating response by invoking a script and communicating with database.

When client sends request for a web page, the web server search for the requested page if requested page is found then it will send it to client with an HTTP response.

If the requested web page is not found, web server will the send an HTTP response: **Error 404 Not found.**

If client has requested for some other resources then the web server will contact to the application server and data store to construct the HTTP response.

Working of web servers



Types of Web Server:

HTTP Server - It handles HTTP request coming from clients browser and transfer the static pages to client in response to their request. This pages runs of the client browser. It generally contains the static pages.

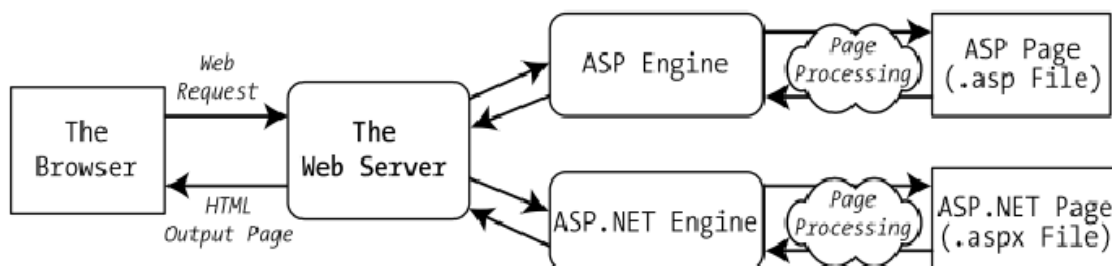
FTP Server - This type of server used for file transfer from one machine (Computer) to another using the internet or intranet. It uses File Transfer Protocols to transfer file from one computer to another. Such type of server uses some file transfer policies, authentication, login validation etc.

Mail Server - A Mail Server store and retrieve mail messages from client mail box.

Application Server - It is installed database and web servers.

Internet Information Server (IIS)

IIS (Internet Information Server) is one of the most powerful web servers from Microsoft that is used to host your ASP.NET Web application. IIS has it's own ASP.NET Process Engine to handle the ASP.NET request. So, when a request comes from client to server, IIS takes that request and process it and send response back to clients.



ASP.NET application

- Introduction

An ASP.NET application is a combination of files, pages, handlers, modules, and executable code that can be invoked from a virtual directory on a web server. Every ASP.NET application is executed inside a separate application domain.

ASP.NET File Types:

Ends with .aspx : These are ASP.NET web pages. They contain the user interface and, optionally, the underlying application code. Users request or navigate directly to one of these pages to start your web application.

Ends with .ascx : These are ASP.NET user controls. User controls are similar to web pages, except that the user can't access these files directly. Instead, they must be hosted inside an ASP.NET web page. User controls allow you to develop a small piece of user interface and reuse it in as many web forms as you want without repetitive code.

Ends with .asmx : These are ASP.NET web services—collections of methods that can be called over the Internet. Web services work differently than web pages, but they still share the same application resources, configuration settings, and memory.

web.config : This is the XML-based configuration file for your ASP.NET application. It includes settings for customizing security, state management, memory management, and much more.

Global.asax : This is the global application file. You can use this file to define global variables

Ends with .vb : These are code-behind files that contain VB code. They allow you to separate the application logic from the user interface of a web page.

Creating ASP.NET page

ASP.NET APPLICATION DIRECTORIES:

Bin : Contains all the compiled .NET components (DLLs) that the ASP.NET web application uses. For example, if you develop a custom component for accessing a database you'll place the component here. ASP.NET will automatically detect the assembly, and any page in the web application will be able to use it.

App_Code : Contains source code files that are dynamically compiled for use in your application. You can use this directory in a similar way as the Bin directory; the only difference is that you place source code files here instead of compiled assemblies.

App_Global : Resources Stores global resources that are accessible to every page in the web application. This directory is used in localization scenarios, when you need to have a website in more than one language.

App_LocalResources : Serves the same purpose as App_GlobalResources, except these resources are accessible to a specific page only.

App_WebReferences: Stores references to web services that the web application uses.

App_Data : Stores data, including SQL Server 2005 Express Edition database files and XML files.

App_Themes : Stores the themes that are used by your web application.

The Page class

IsPostBack : This Boolean property indicates whether this is the first time the page is being run (False) or whether the page is being resubmitted in response to a control event, typically with stored view state information(True). You'll usually check this property in the Page.Load event handler to ensure that your initial web page initialization is only performed once.

EnableViewState : When set to False, this overrides the EnableViewState property of the contained controls, thereby ensuring that no controls will maintain state information.

Application : This collection holds information that's shared between all users in your website. For example, you can use the Application collection to count the number of times a page has been visited.

Session : This collection holds information for a single user, so it can be used in different pages. For example, you can use the Session collection to store the items in the current user's shopping basket on an e-commerce website.

Cache : This collection allows you to store objects that are time- consuming to create so they can be reused in other pages or for other clients. This technique, when implemented properly, can improve performance of your web pages.

Request : This refers to an HttpRequest object that contains information about the current web request.

Response : This refers to an HttpResponse object that represents the response ASP.NET will send to the user's browser.

Server: This refers to an HttpServerUtility object that allows you to perform a few miscellaneous tasks. For example, it allows you to encode text so that it's safe to place it in a URL or in the HTML markup of your page.

User : If the user has been authenticated, this property will be initialized with user information.

Web.config File

```
<?xml version="1.0" ?>
<configuration>
<configSections>...</configSections>
<appSettings>...</appSettings>
<connectionStrings>...</connectionStrings>
<system.web>...</system.web>
```

```
<system.codedom>...</system.codedom>  
<system.webServer>...</system.webServer>  
</configuration>
```

The three sections in the web.config file are:

The <appSettings> section allows you to add your own miscellaneous pieces of information.

The <connectionStrings> section allows you to define the connection information for accessing a database.

The <system.web> section holds every ASP.NET setting you'll need to configure. Inside the <system.web> element are separate elements for each aspect of website configuration.